

DDS Frequency Generator

PCB Design Project | Tool: KiCad | 2-Layer PCB

Design Tool	KiCad
Board Type	2-Layer PCB
Power Supply	+12V, -12V, +5V

1. Project Overview

DDS-based frequency generator PCB using AD9833 + ATmega328A. Generates sine/triangle/square waves controlled via SPI. Includes LM318M output amplifier, 24MHz reference oscillator, and coaxial output. Designed in KiCad.

2. Key Components

- AD9833BRMZ-REEL – DDS signal generator IC
- ATmega328-A – Microcontroller
- CFPS-72 24MHz Crystal Oscillator
- AMS1117-5.0 – 5V LDO Regulator
- LM318M – High-speed Op-Amp
- Potentiometers (2x) – Frequency/Amplitude adjust
- SW_SPDT – Waveform select switch
- Coaxial Output Connector

3. Design Features

- AD9833 generates sine, triangle, square waveforms
- ATmega328A controls frequency via SPI interface
- 24MHz CFPS-72 reference oscillator for precision
- LM318M output amplifier for signal conditioning
- Dual 12V supply for op-amp headroom
- Potentiometers for frequency fine-tuning

4. Signal / Functional Flow

- [1] ATmega328A – MCU Controller ↓
- [2] SPI Interface ↓
- [3] AD9833 – DDS Waveform Generator ↓
- [4] LM318M – Output Amplifier ↓
- [5] Waveform Select Switch (SW_SPDT) ↓
- [6] Coaxial Output Connector ↓
- [7] AMS1117-5.0 ← 5V Supply ↓
- [8] ±12V Supply ← Dual Rail Power Input

5. Resume / Portfolio Description

- Designed DDS Frequency Generator PCB in KiCad using AD9833 waveform IC controlled by ATmega328A over SPI with 24MHz reference crystal oscillator.

- Included LM318M high-speed op-amp output stage, dual $\pm 12\text{V}$ supply, and coaxial RF output connector with waveform select switch.

6. Short Description (Portfolio / GitHub – ≤ 350 chars)

DDS-based frequency generator PCB using AD9833 + ATmega328A. Generates sine/triangle/square waves controlled via SPI. Includes LM318M output amplifier, 24MHz reference oscillator, and coaxial output. Designed in KiCad.

(218 characters)